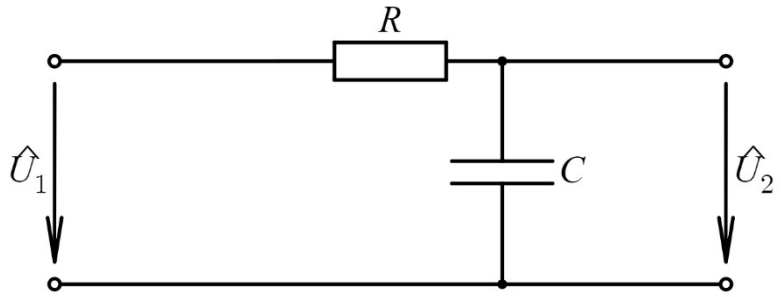
# EK - MO 11. Frekvenční charakteristiky pasivních dvojbranů

## A) RC

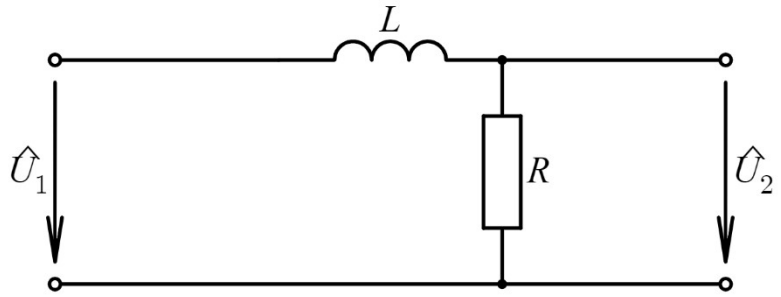



$$\widehat{U_2} = \widehat{U_1} \cdot \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \widehat{U_1} \cdot \frac{1}{1 + j\omega RC}$$

$$\tau_0 = RC \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{RC} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC}$$

$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}} = \frac{\widehat{U_1} \cdot \frac{1}{1 + j\omega RC}}{\widehat{U_1}} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\omega\tau_0}$$

## B) LR




$$\widehat{U_2} = \widehat{U_1} \cdot \frac{R}{R + j\omega L} = \widehat{U_1} \cdot \frac{R}{R\left(1 + j\omega \frac{L}{R}\right)} = \widehat{U_1} \cdot \frac{1}{1 + j\omega \frac{L}{R}}$$

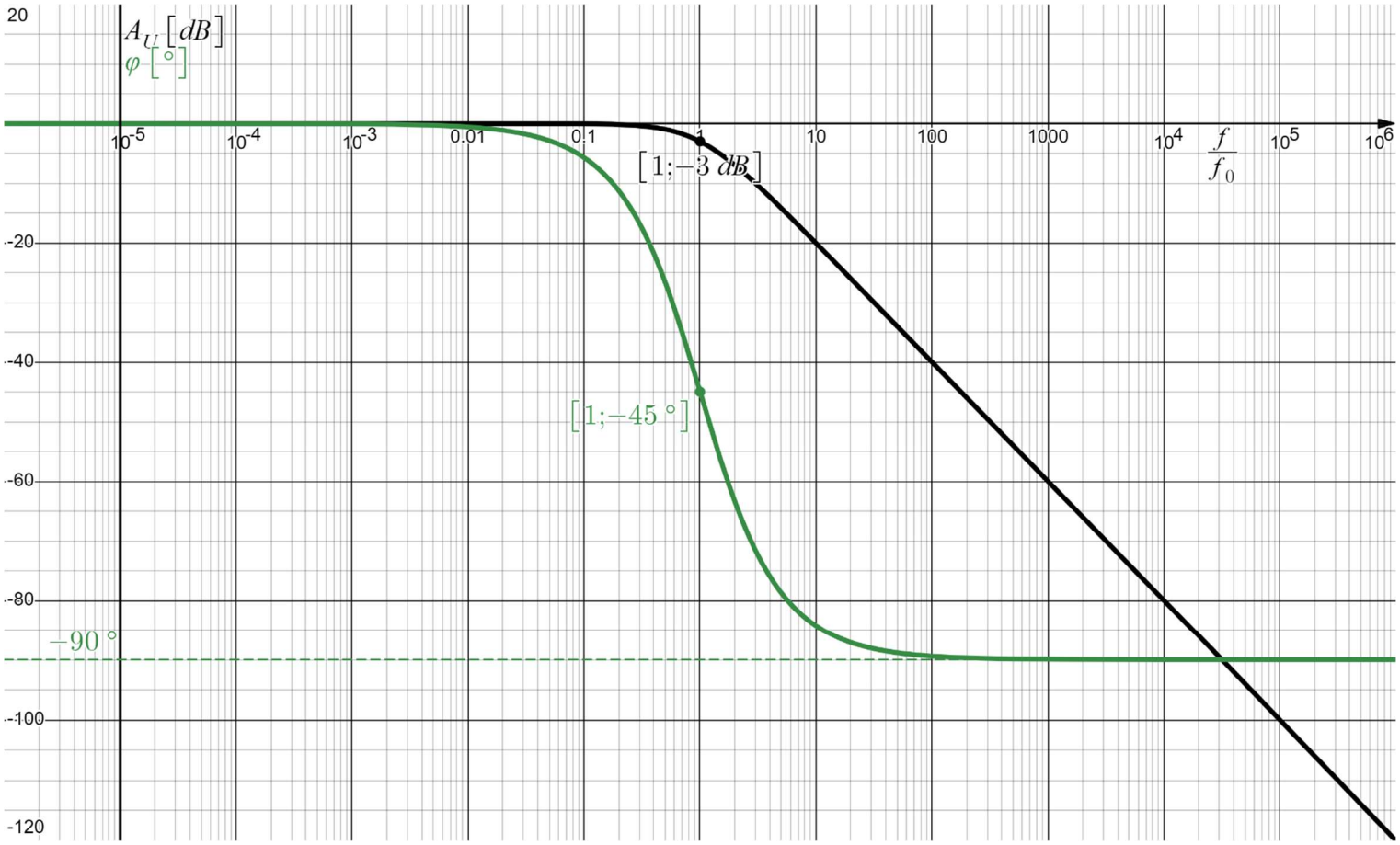
$$\tau_0 = \frac{L}{R} \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{\frac{L}{R}} = \frac{R}{L} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{R}{2\pi L}$$

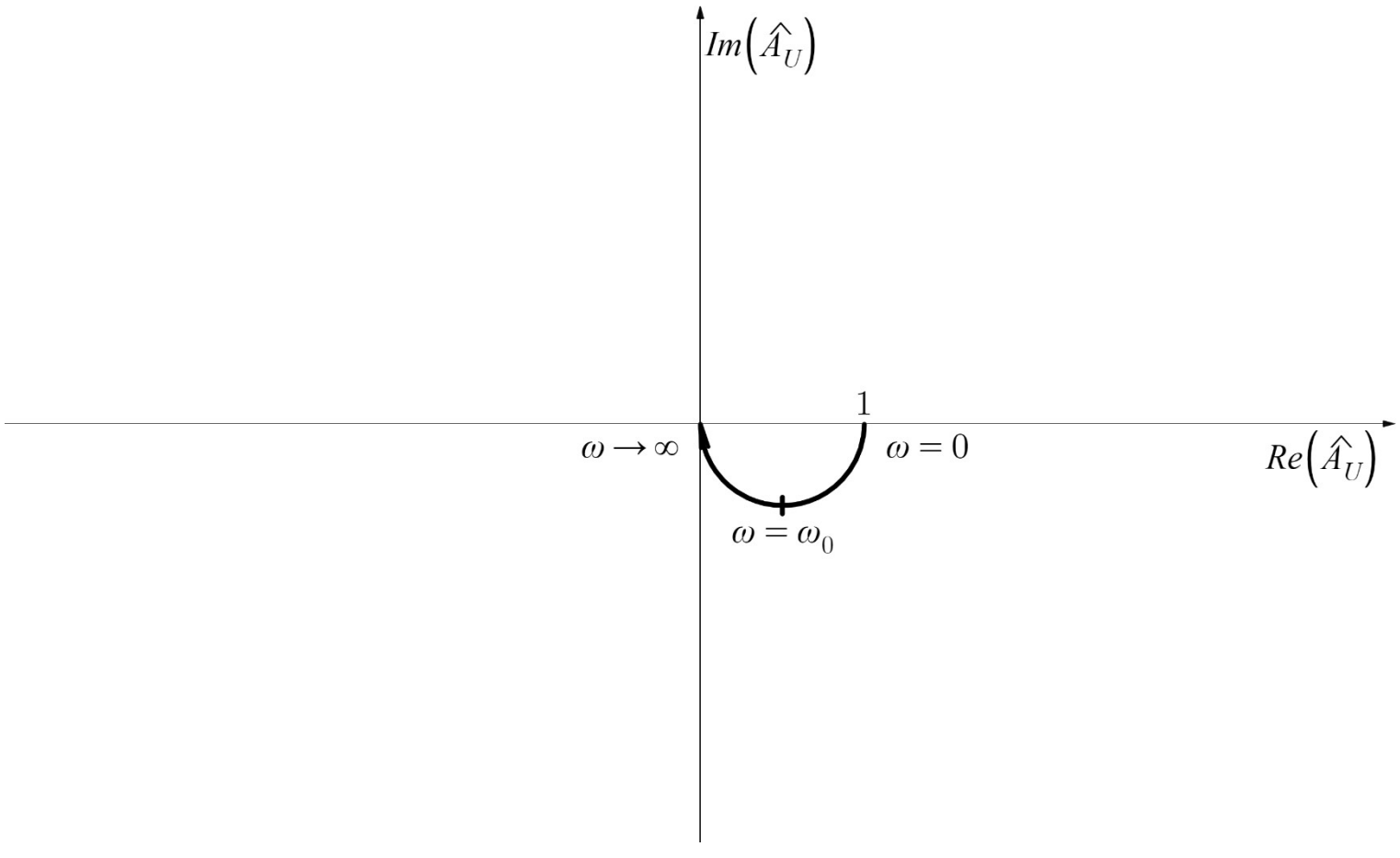
$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}} = \frac{\widehat{U_1} \cdot \frac{1}{1 + j\omega \frac{L}{R}}}{\widehat{U_1}} = \frac{1}{1 + j\omega \frac{L}{R}} = \frac{1}{1 + j\omega\tau_0}$$

$$A_{U\,dB} = 20 \log \left| \frac{\widehat{U_2}}{\widehat{U_1}} \right| = 20 \log \left| \frac{1}{1 + j\omega\tau_0} \right|$$

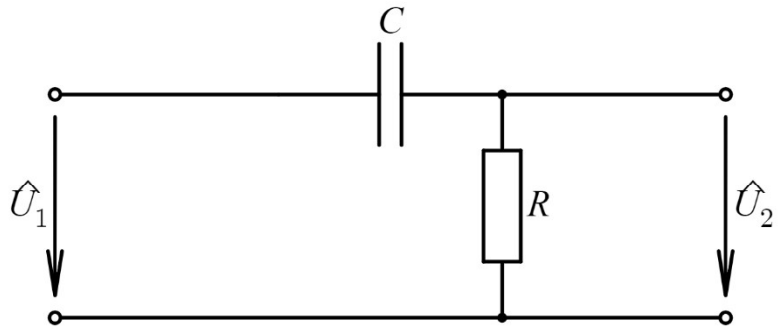
Lze vidět, že RC a LR články mají stejný přenos $\left(\frac{1}{1+j\omega\tau_0}\right)$ takže na venek se chovají identicky (pokud předpokládáme ideální součástky).

Těmto dvěma článkům říkáme *integrační*.



## C)CR



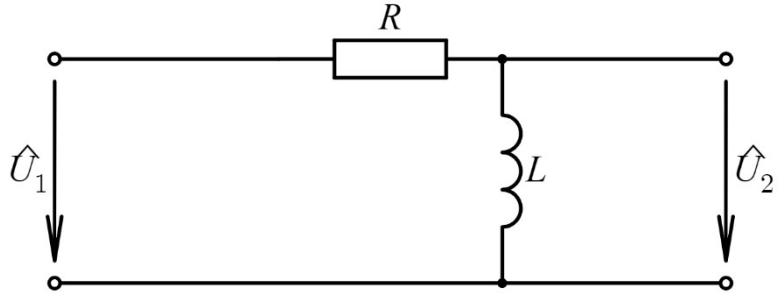

$$\widehat{U_2} = \widehat{U_1} \cdot \frac{R}{R + \frac{1}{j\omega C}} = \widehat{U_1} \cdot \frac{j\omega RC}{1 + j\omega RC}$$

$$\tau_0 = RC \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{RC} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi RC}$$

$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}} = \frac{\widehat{U_1} \cdot \frac{j\omega RC}{1 + j\omega RC}}{\widehat{U_1}} = \frac{j\omega RC}{1 + j\omega RC} = \frac{j\omega\tau_0}{1 + j\omega\tau_0}$$

$$A_{U\,dB} = 20 \log \left| \frac{\widehat{U_2}}{\widehat{U_1}} \right| = 20 \log \left| \frac{j\omega\tau_0}{1 + j\omega\tau_0} \right|$$

## D)RL




$$\widehat{U_2} = \widehat{U_1} \cdot \frac{j\omega L}{R + j\omega L} = \widehat{U_1} \cdot \frac{j\omega L}{R\left(1 + j\omega \frac{L}{R}\right)} = \widehat{U_1} \cdot \frac{j\omega \frac{L}{R}}{1 + j\omega \frac{L}{R}}$$

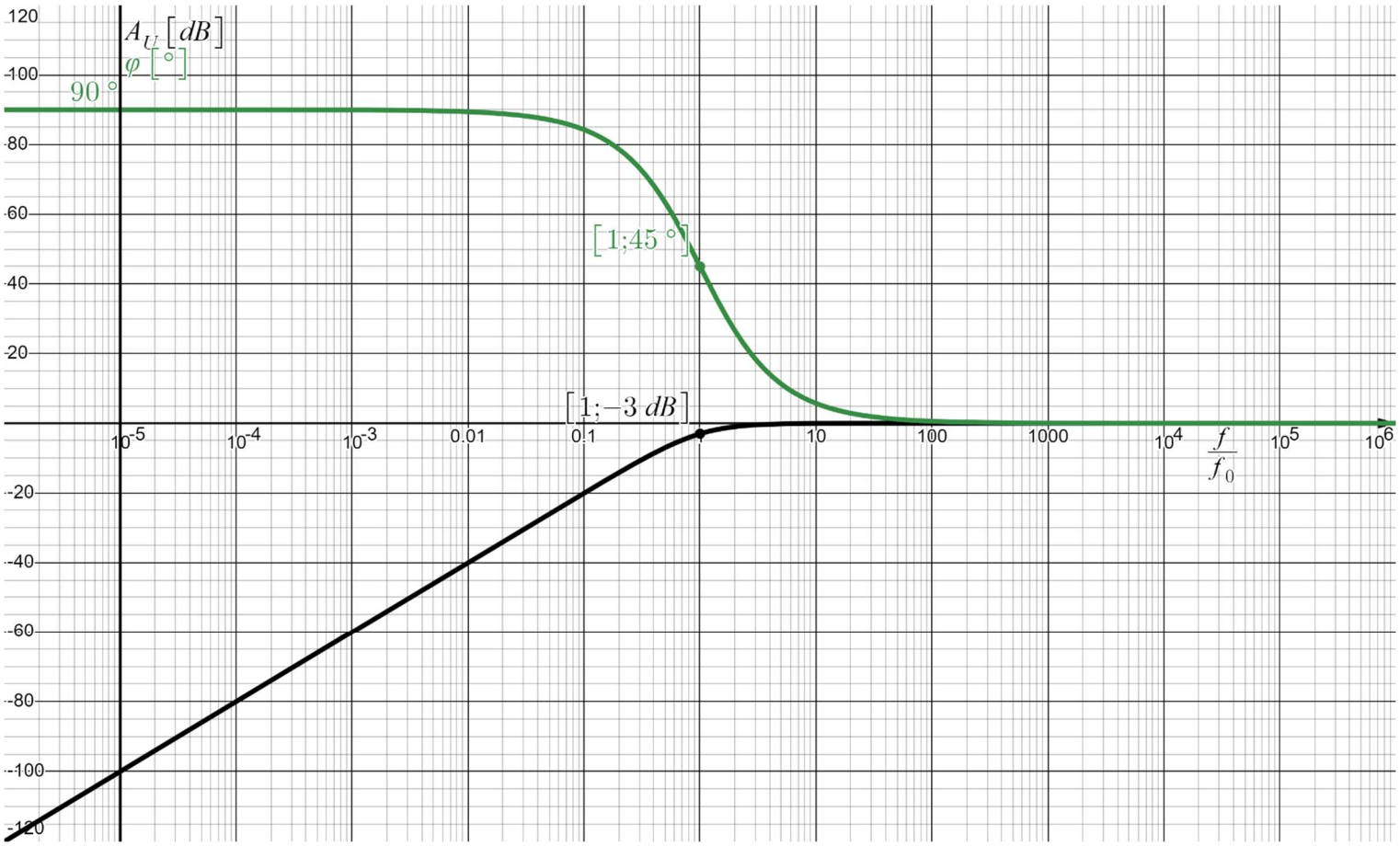
$$\tau_0 = \frac{L}{R} \implies \omega_0 = \frac{1}{\tau_0} = \frac{1}{\frac{L}{R}} = \frac{R}{L} \implies f_0 = \frac{\omega_0}{2\pi} = \frac{R}{2\pi L}$$

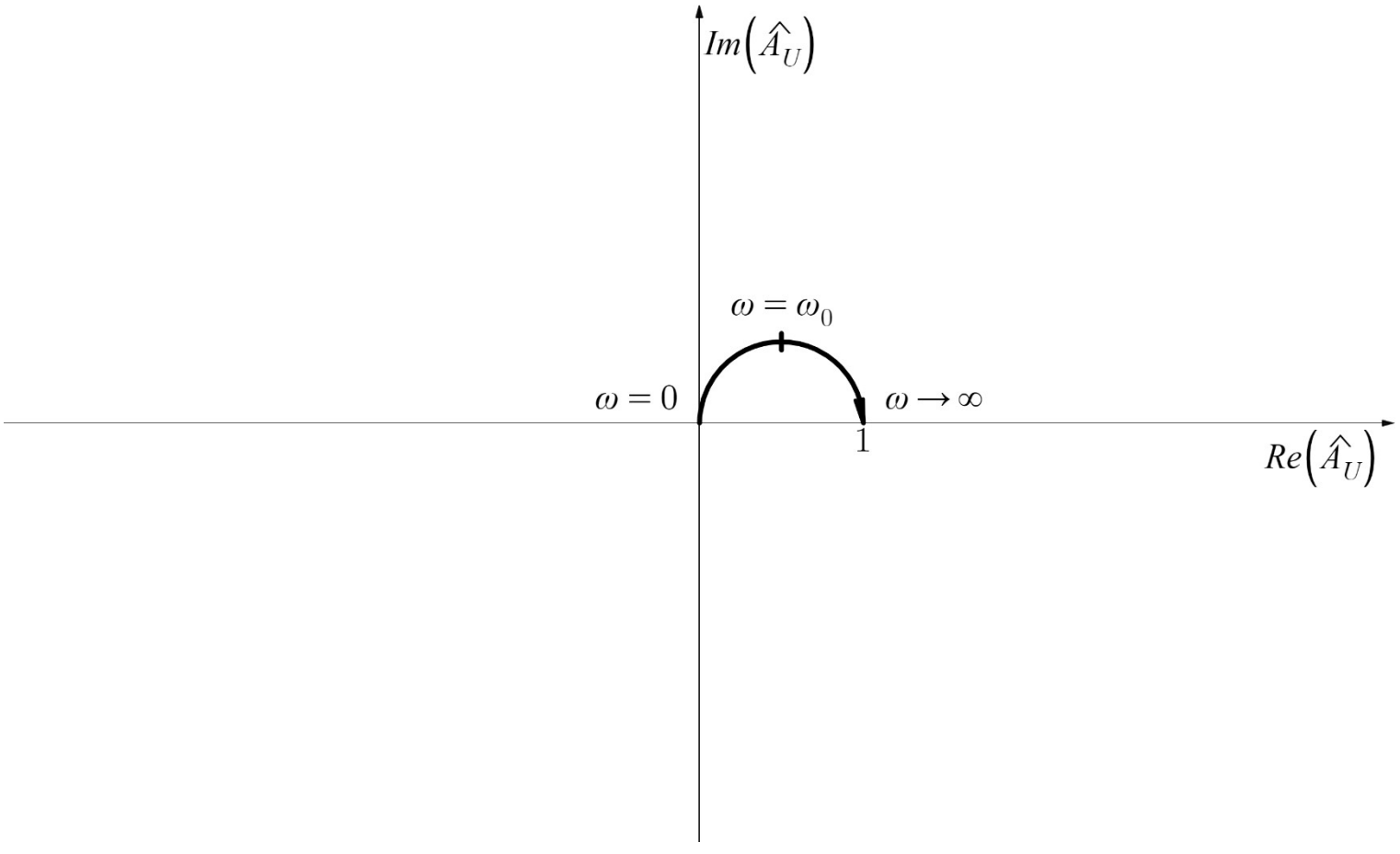
$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}} = \frac{\widehat{U_1} \cdot \frac{j\omega \frac{L}{R}}{1 + j\omega \frac{L}{R}}}{\widehat{U_1}} = \frac{j\omega \frac{L}{R}}{1 + j\omega \frac{L}{R}} = \frac{j\omega\tau_0}{1 + j\omega\tau_0}$$

$$A_{U\,dB} = 20 \log \left| \frac{\widehat{U_2}}{\widehat{U_1}} \right| = 20 \log \left| \frac{j\omega\tau_0}{1 + j\omega\tau_0} \right|$$

Lze vidět, že RC a LR články mají stejný přenos $\left(\frac{j\omega\tau_0}{1+j\omega\tau_0}\right)$ takže na venek se chovají identicky (pokud předpokládáme ideální součástky).

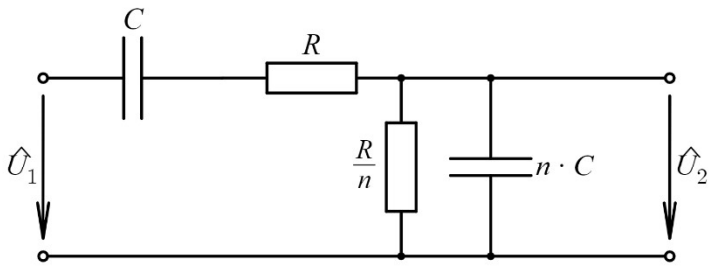
Těmto dvěma článkům říkáme *derivační*.

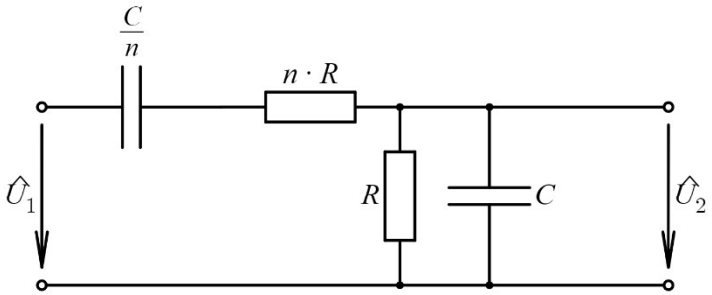


## E) Wienův člen

Často se používá se součástkami $\frac{R_1}{R_2} = \frac{C_2}{C_1} = n$, ale nejčastěji $R_1 = R_2$ a $C_1 = C_2$.

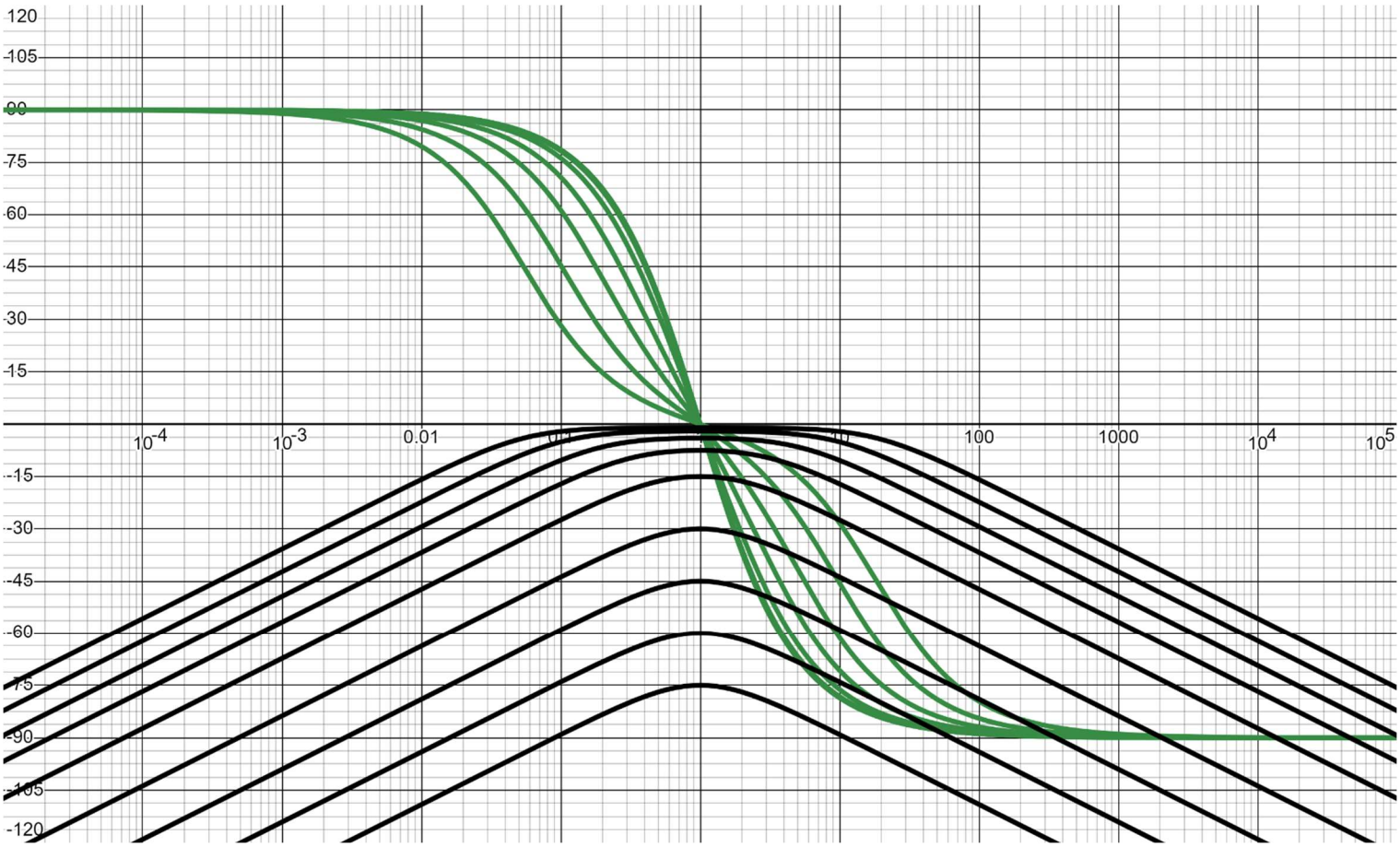


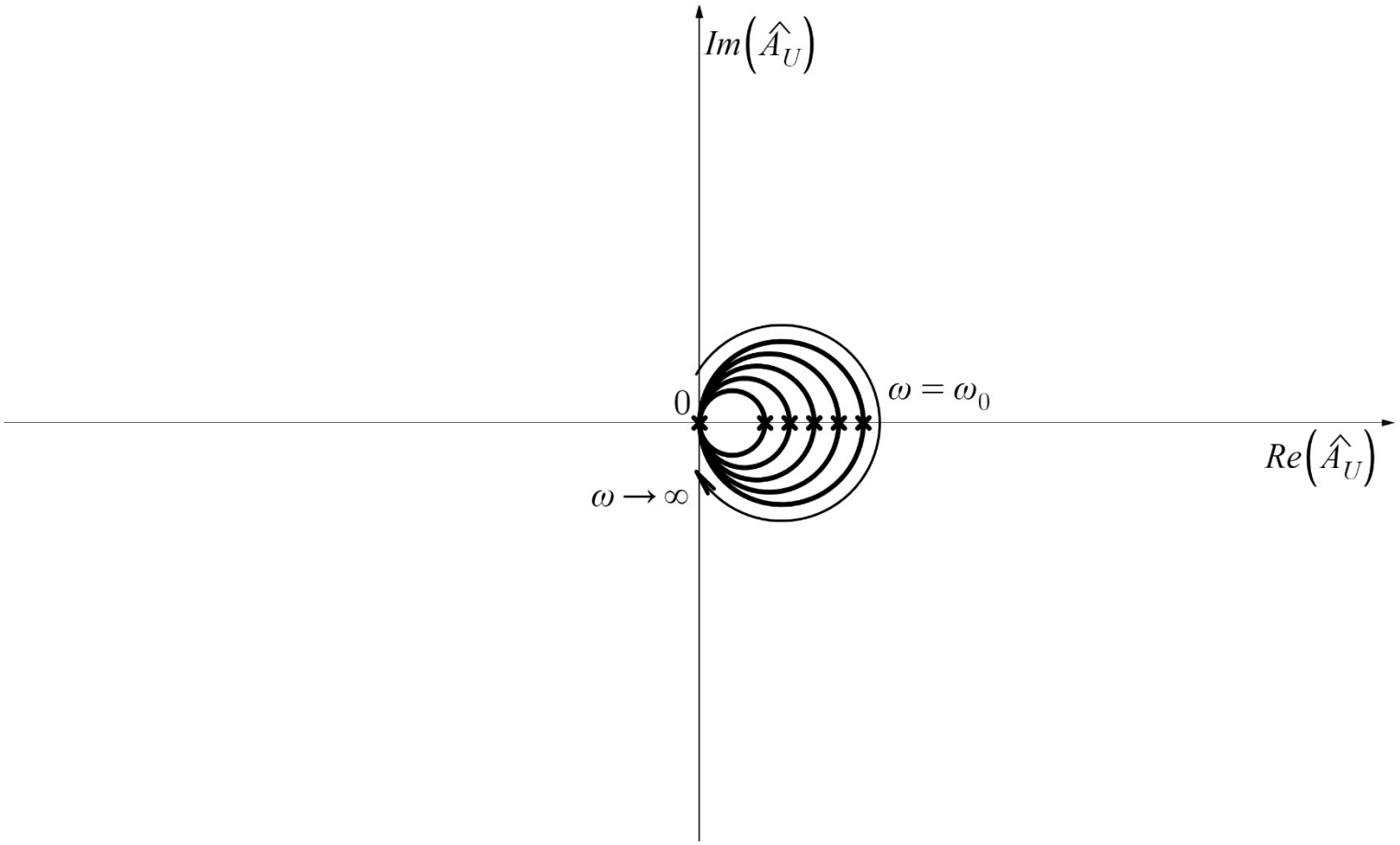




$$f_0 = \frac{1}{2\pi\sqrt{R_1 C_1 R_2 C_2}} \overset{!}{=} \frac{1}{2\pi\sqrt{RC\frac{R}{n}nC}} = \frac{1}{2\pi\sqrt{R^2C^2}} = \frac{1}{2\pi RC}$$

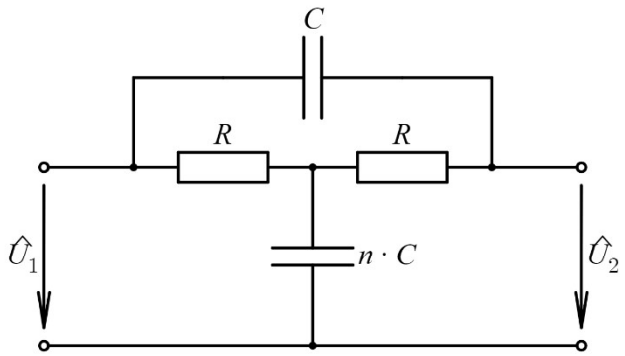
$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}}, A_0 = \widehat{A_U}\big|_{f=f_0} = \frac{1}{1 + \frac{R_1}{R_2} + \frac{C_2}{C_1}} \overset{!}{=} \frac{1}{1 + \frac{nR}{R} + \frac{nC}{C}} = \frac{1}{1+2n} \overset{n=1}{=} \frac{1}{3}$$

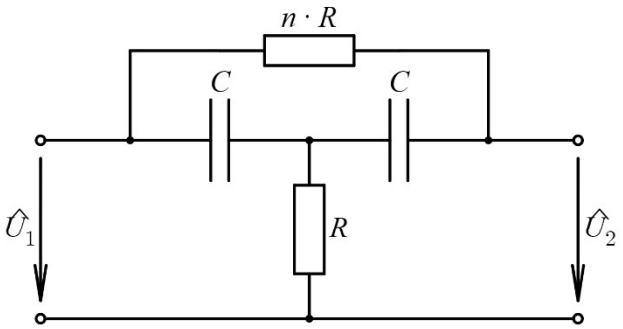
$$B = \left(2 + \frac{1}{n}\right) f_0$$





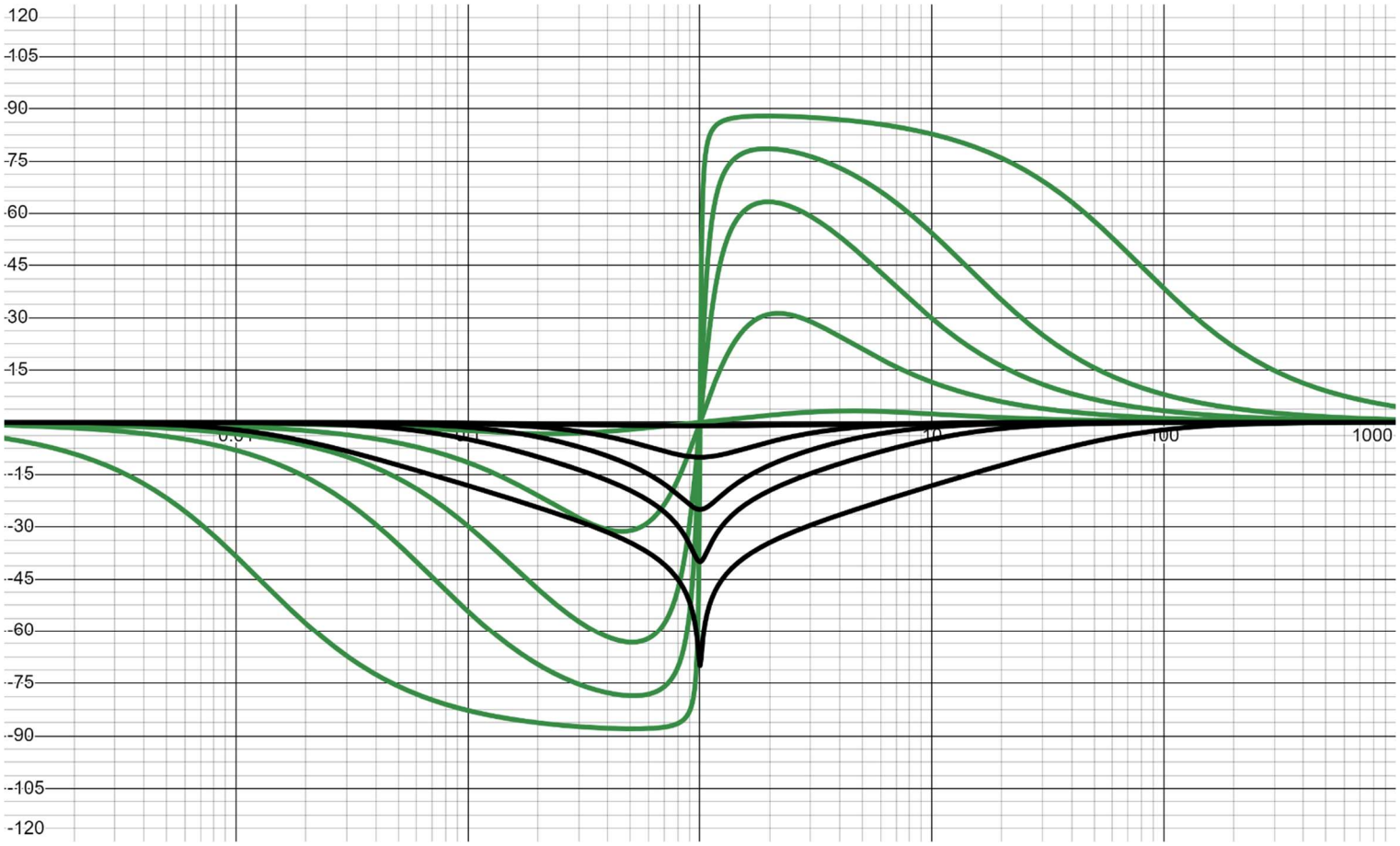

## F) Přemostěný T-článek

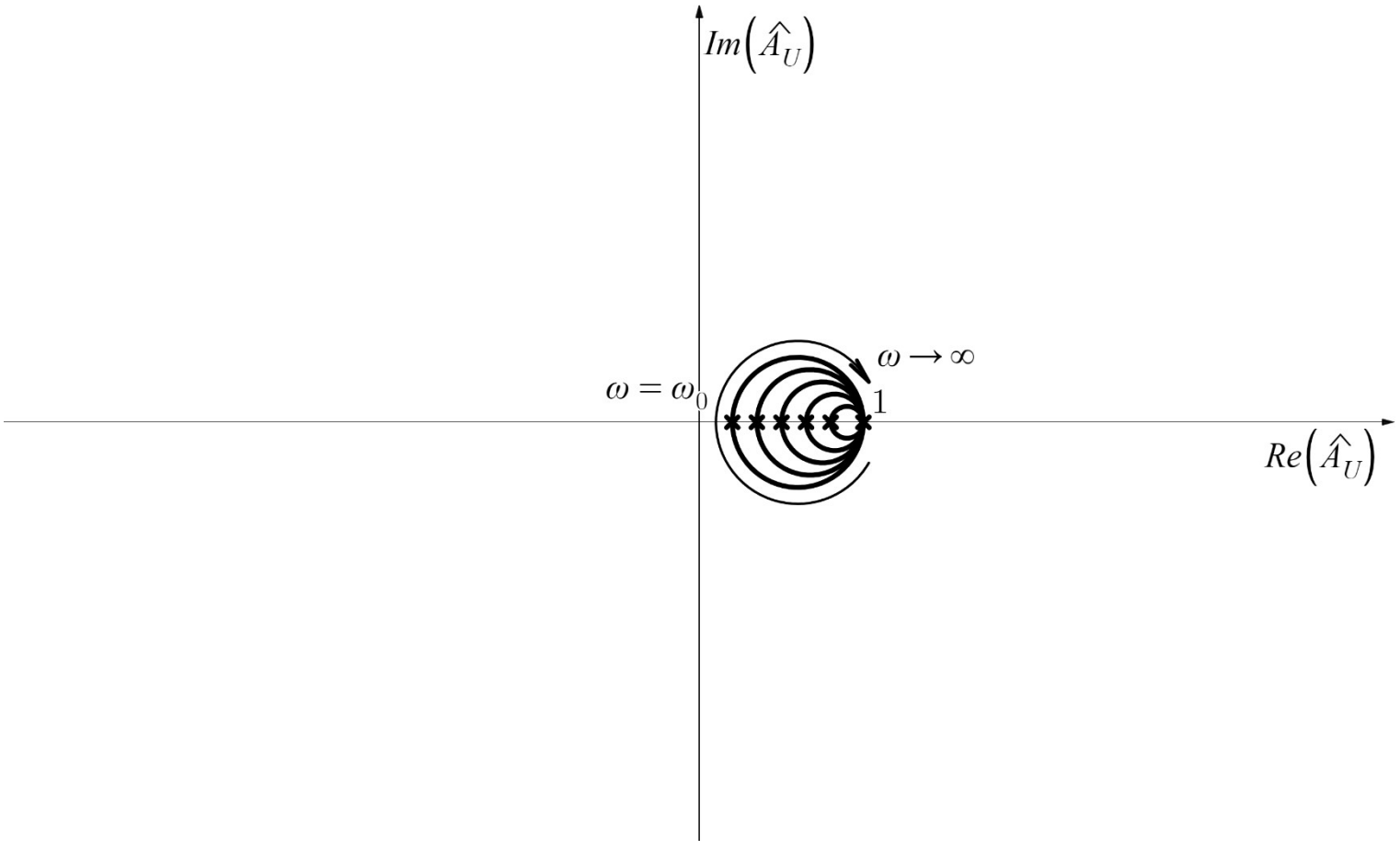



$$f_0 = \frac{1}{2\pi RC}$$

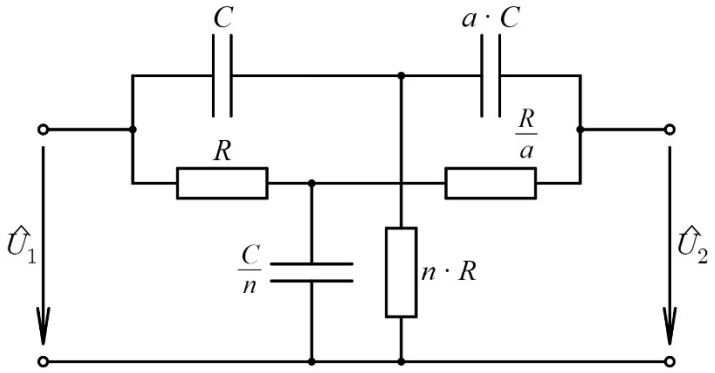
$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}} = \frac{F\sqrt{n} - 2j}{F\sqrt{n} - (2+n)j}, A_0 = \widehat{A_U}\Big|_{f=f_0} \overset{!}{=} \frac{2}{2+n}$$


120
105
90
75
60
45
30
15
-15
-30
-45
-60
-75
-90
-105
-120
10
100
1000

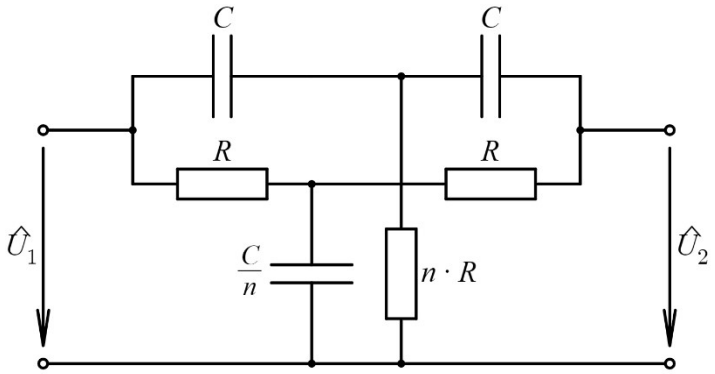

$Im\left(\hat{A}_U\right)$
$\omega \to \infty$
$\omega = \omega_0$
1
$Re\left(\hat{A}_U\right)$

## G) Dvojitý T-článek

Nejobecnější Dvojitý T-článek je ve tvaru:




Nejčastěji se ale používá pro $a = 1$:




$$f_0 = \frac{1}{2\pi RC}$$

$$\widehat{A_U} = \frac{\widehat{U_2}}{\widehat{U_1}} = \frac{n - \omega^2 C^2 R^2 n + j\omega(CRan^2 + CRn^2 - CRn)}{n - \omega^2 C^2 R^2 n + j\omega(CRan^2 + CRan + CRn^2 + CR)}$$

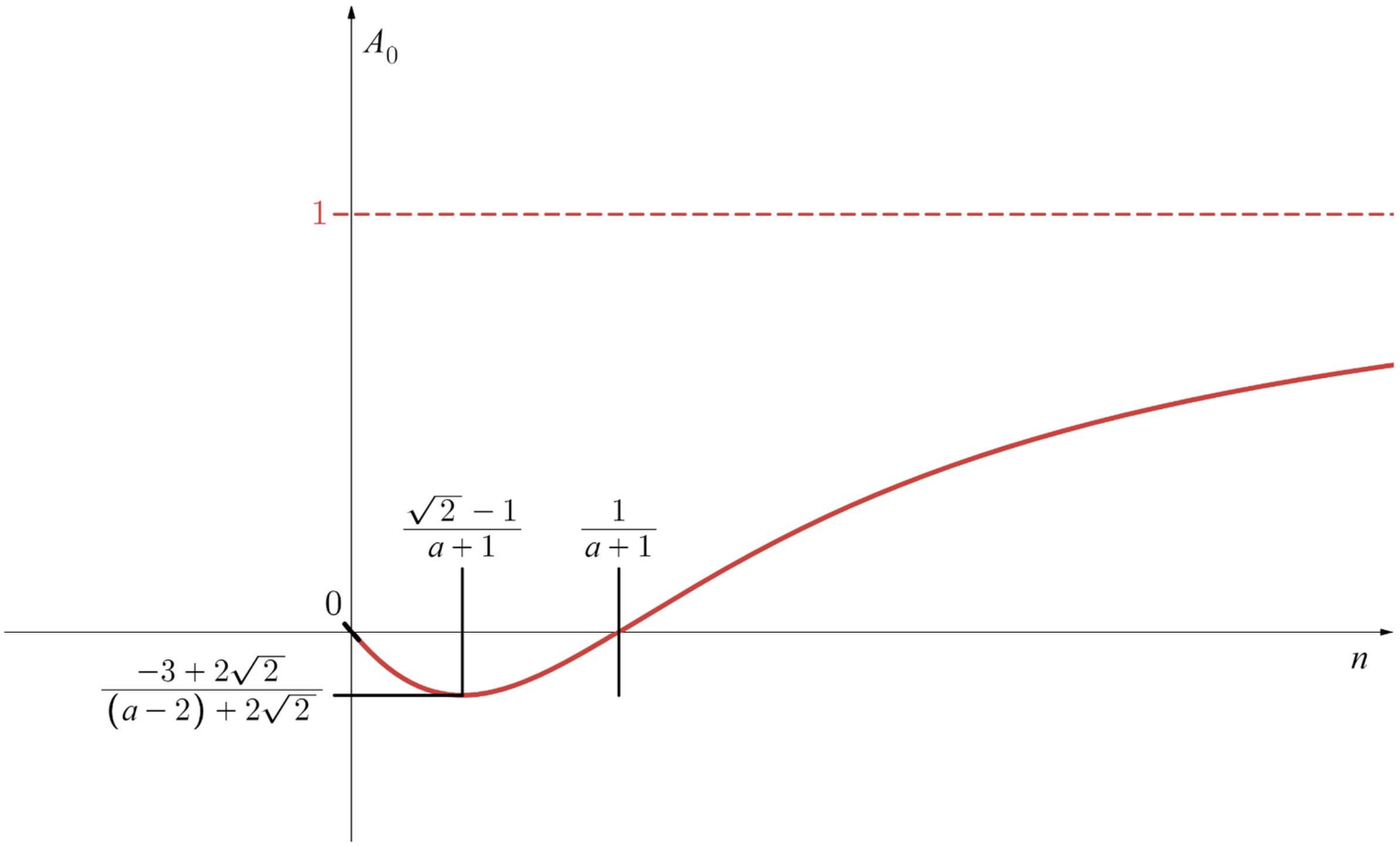
$$\overset{a=1}{=} \frac{n - \omega^2 C^2 R^2 n + j\omega(2CRn^2 - CRn)}{n - \omega^2 C^2 R^2 n + j\omega(2CRn^2 + CRn + CR)}$$

$$A_0 = \widehat{A_U}\big|_{f=f_0} = \frac{(a+1)n^2 - n}{(a+1)n^2 + an + 1} \overset{a=1}{=} \frac{2n^2 - n}{2n^2 + n + 1}$$

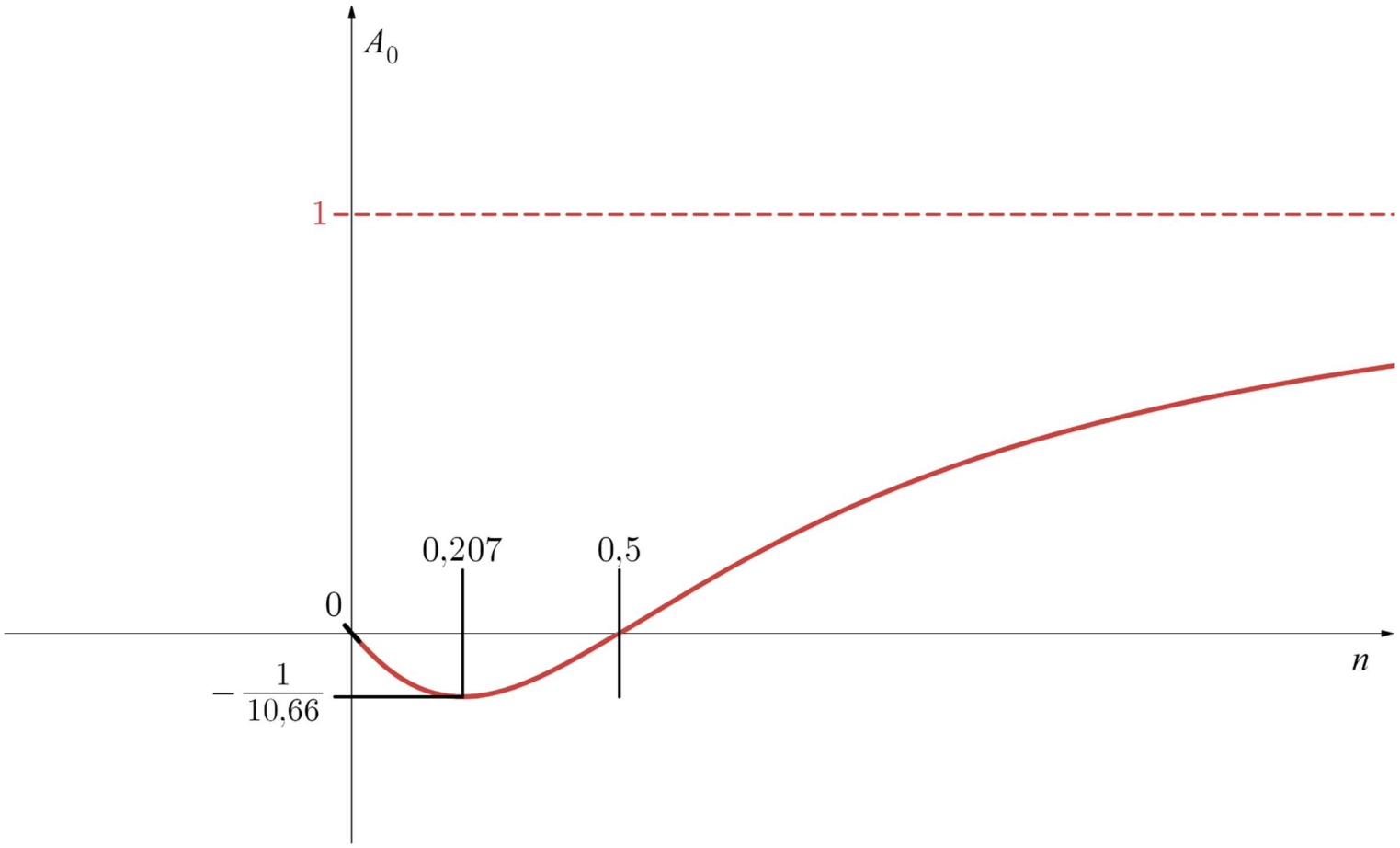
$$A_0 = \frac{(a+1)n^2 - n}{(a+1)n^2 + an + 1} = 0 \implies n = \frac{1}{a+1} \overset{a=1}{=} \frac{1}{2}$$

$$\min(A_0) = \frac{-3 + 2\sqrt{2}}{(a-2) + 2\sqrt{2}} \overset{a=1}{\approx} -\frac{1}{10{,}66} \text{ pro } n = \frac{\sqrt{2} - 1}{a+1} \overset{a=1}{\approx} 0{,}207$$

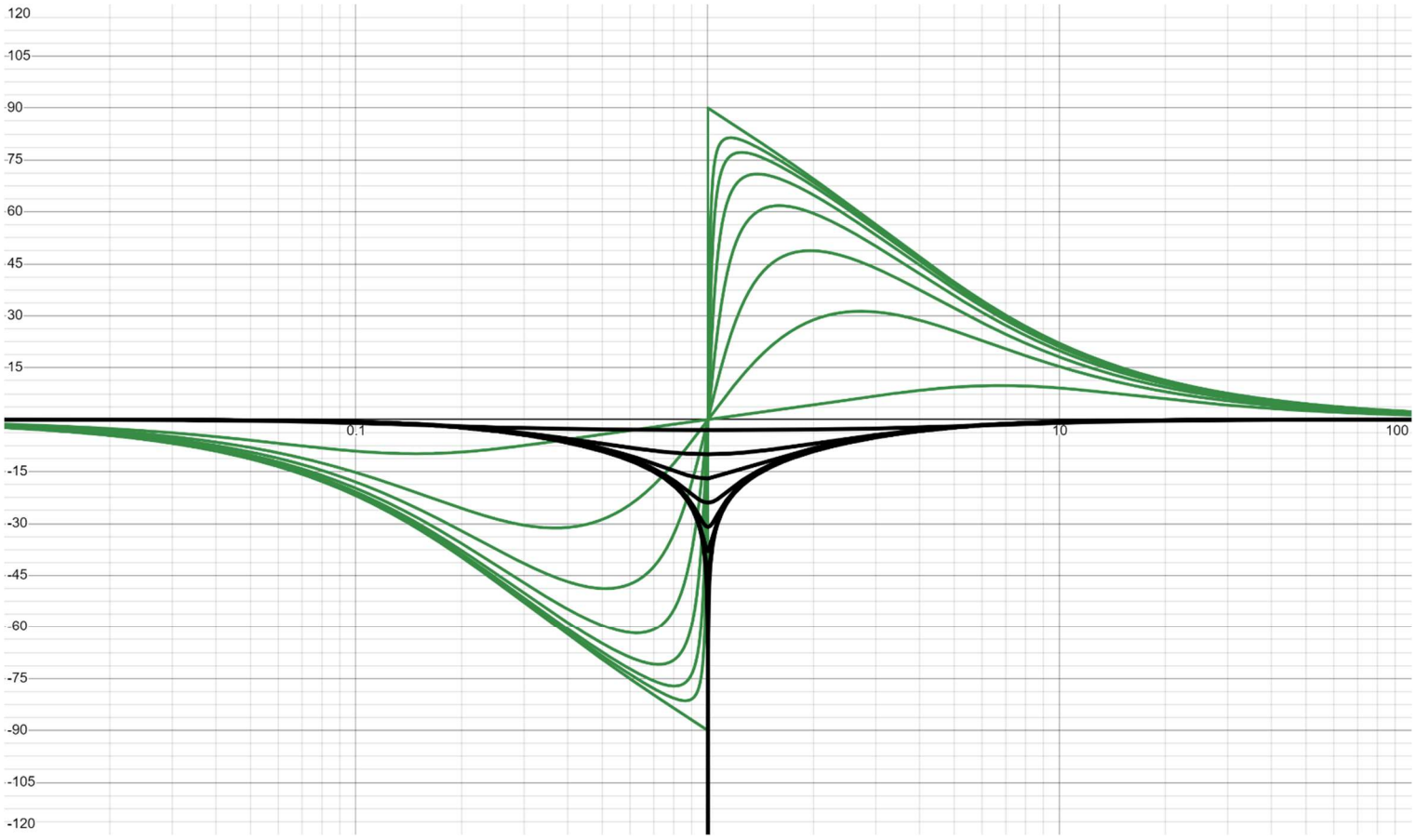
$a > 0$


$A_0$
1
$\frac{\sqrt{2}-1}{a+1}$
$\frac{1}{a+1}$
0
$\frac{-3+2\sqrt{2}}{(a-2)+2\sqrt{2}}$
$n$

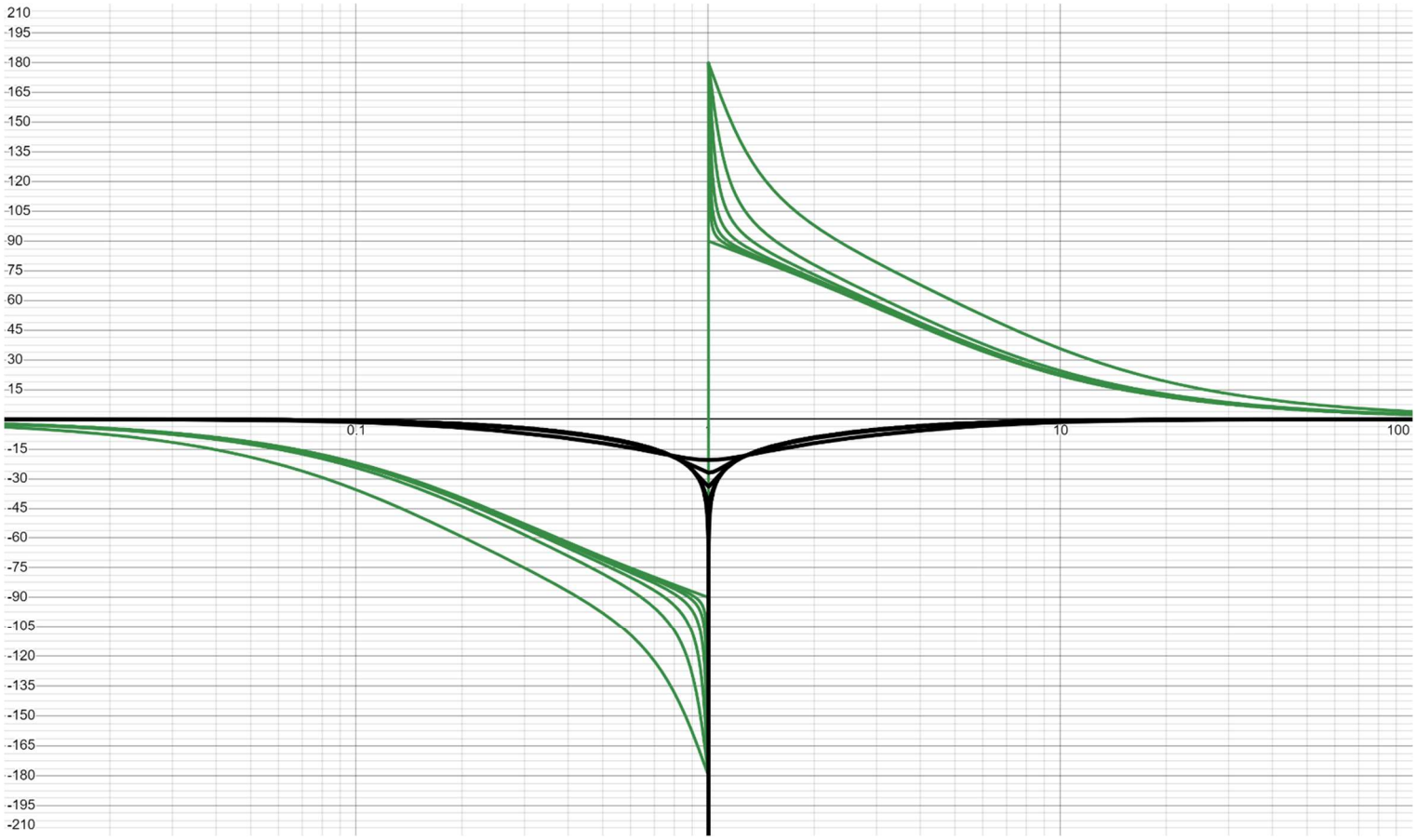
$a = 1$

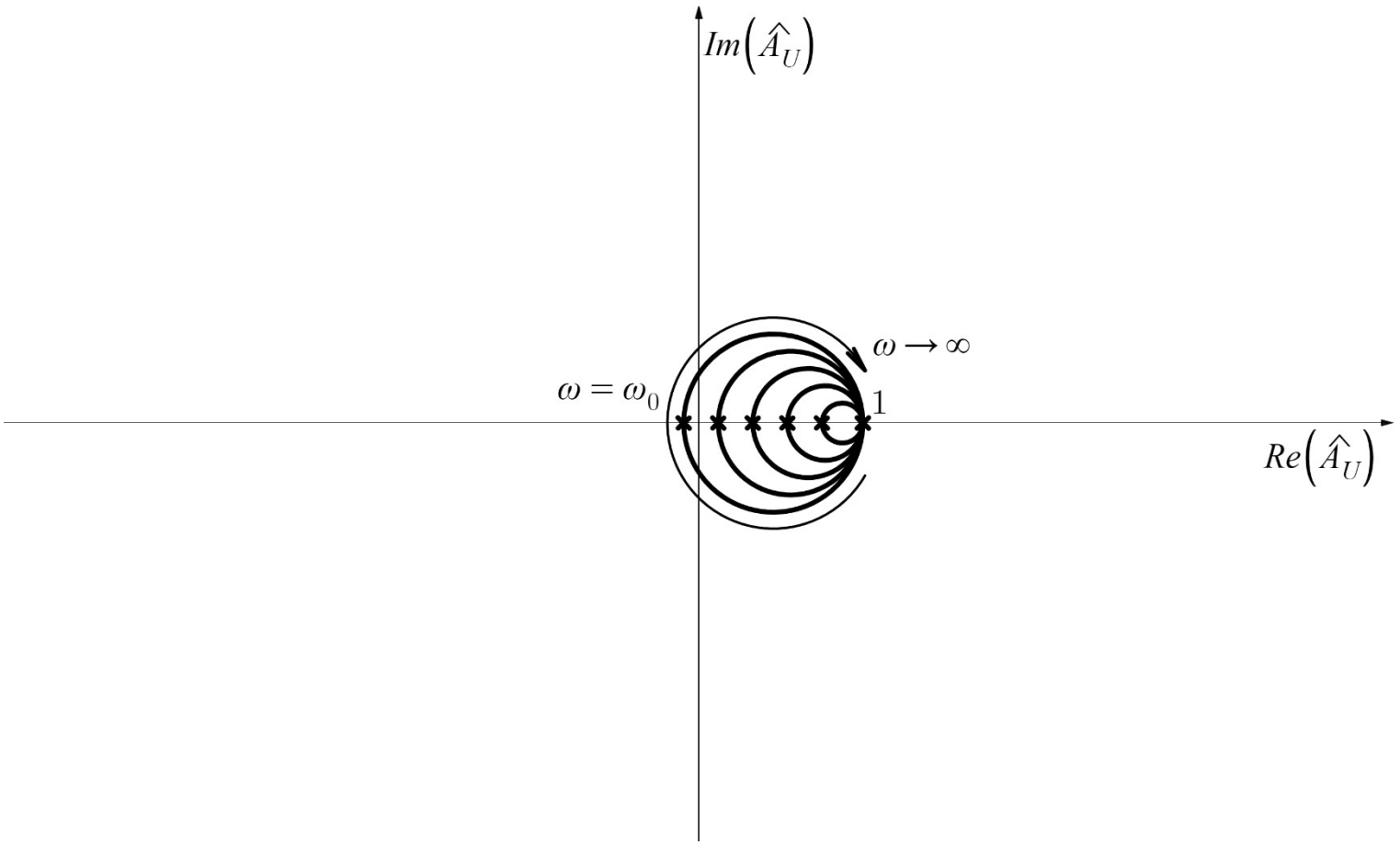

$A_0$
1
0,207
0,5
0
$-\frac{1}{10,66}$
$n$

$$n \geq \frac{1}{a+1} \overset{a=1}{=} \frac{1}{2}$$




$$0 < n \leq \frac{1}{a+1} \overset{a=1}{=} \frac{1}{2}$$




$Im\left(\widehat{A}_U\right)$
$\omega \to \infty$
$\omega = \omega_0$
1
$Re\left(\widehat{A}_U\right)$